Mahindra University
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ÉCOLE CENTRALE SCHOOL OF ENGINEERING

Mathematics-1 (MA 1101)
Tutorial Sheet-3
Topics: Sequences, Series and Ordinary Differential Equations

***Problems to be worked out in tutorial**

†Assignment Problems

1. Does the following sequences converge? Find the limit of each convergent sequence.
   (a) $u_n = \frac{n}{n+1} - \frac{n+1}{n}$
   (b) † $u_n = \frac{1+(-1)^n}{n}$
   (c) * $u_n = \sqrt{n+1} - \sqrt{n}$
   (d) † $u_n = e^{-i\frac{n\pi}{2}}$

2. Determine the convergence of the following series. In case of convergence, find its value (wherever possible).
   (a) $\sum_{n=1}^{\infty} \frac{2}{3^{n-1}}$
   (b) † $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$
   (c) * $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2}$
   (d) † $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-\sqrt{n}}{\sqrt{n^2+n}}$

3. Determine the convergence or divergence of the given series. In case of convergence, determine whether the series converges absolutely or conditionally.
   (a) * $\sum_{n=1}^{\infty} \frac{(-1)^{n(n-1)/2}}{2^n}$
   (b) † $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[n]{n}}$
   (c) † $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n^2+1}$

4. Find the curve passing through the point $(1, 0)$ and having at each of its points the slope $-\frac{x}{y}$

5. Solve the following Initial Value Problems (IVP)

   (a) * $\frac{dy}{dx} = 2y + e^{2x}$, $y(0) = 3$

   (b) † $\frac{dy}{dx} = \frac{2}{x}y + x$, $y(1) = 2$

6. Define the Wronskian $w(y_1, y_2)$ of any two differentiable functions y_1 and y_2 defined in an interval $(a, b) \subset R$. Show that $w(y_1, y_2) = 0$ if y_1 and y_2 are linearly dependent.

7. Find the general solution of the following second order equations using the given known solution y_1.

   (a) * $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$ where $y_1(x) = x$.

   (b) † $x\frac{d^2y}{dx^2} - (2x+1)\frac{dy}{dx} + (x+1)y = 0$ where $y_1(x) = e^x$.

8. Solve the following using the method of variation of parameters ($D^n \equiv \frac{d^n}{dx^n}$)

   (a) † $(D^2 + 1)y = cosecx$

   (b) $(D^2 - D - 6)y = e^{-x}$

   (c) † $(D^2 + a^2)y = \tan ax$

   (d) * $x^2y'' - 4xy' + 6y = 21x^{-4}$